Final project presentation

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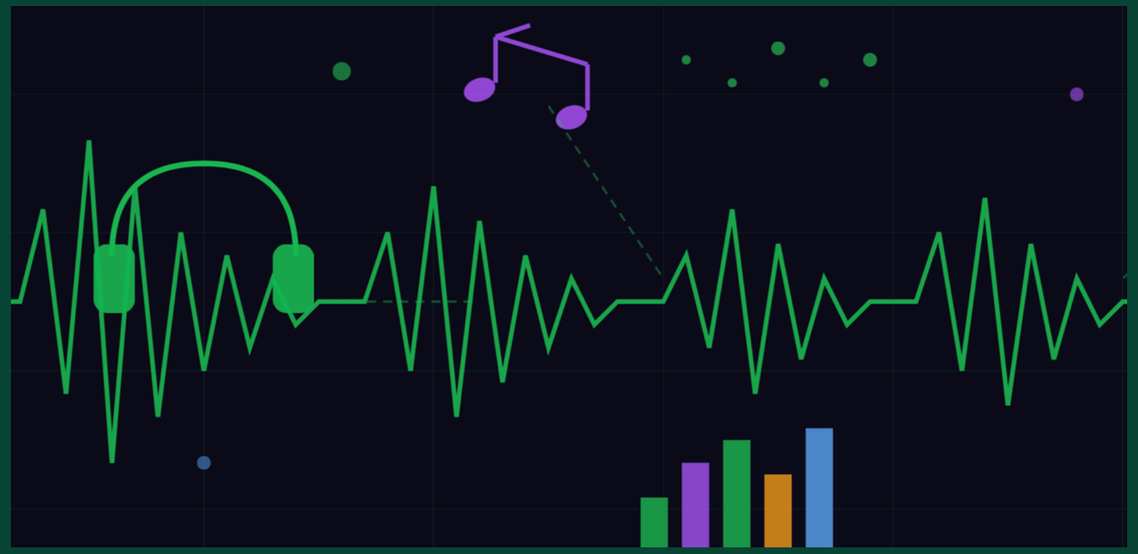
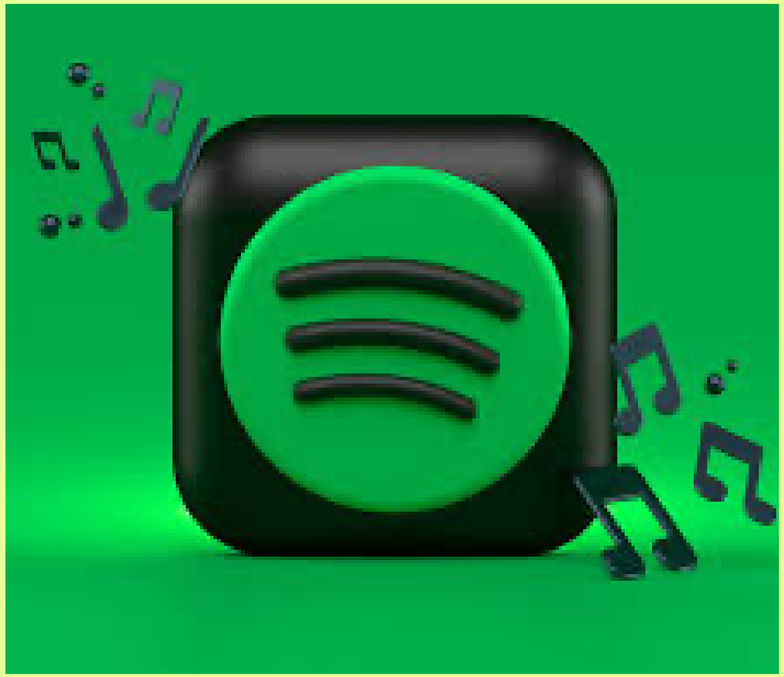


Predicting the Popularity of Hip-Hop Songs Using Audio Features

DSCI-510PRINCIPLES OF PROGRAMMING FOR DATA SCIENCE
26 SPRING

Introduction
Topic： Music

Can audio features predict whether a hip-hop song will be commercially popular?



Definition of “Hit”:

- A song is labeled as a hit if its Spotify popularity score ⩾ 50

Data & Features:

- Spotify API + Kaggle supplementary datasets
- Some Key features: danceability, energy, tempo, loudness, valence

Method:

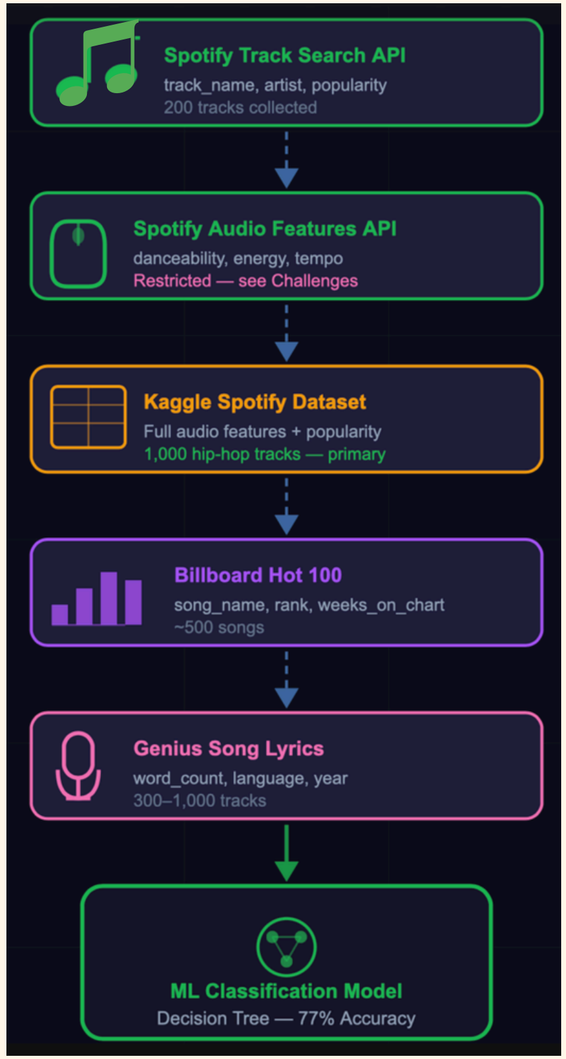
- Exploratory Data Analysis (EDA)
- Classification models: Logistic Regression & Decision Tree

Goal:

- Identify which audio features are associated with popularity and evaluate their predictive power

Data Sources

Spotify Track Search API	https://developer.spotify.com/documentation/web-api	CSV	track metadata: name, artist, popularity, release date, explicit flag
Spotify Audio Features API	https://developer.spotify.com/documentation/web-api	CSV	danceability, energy, tempo, loudness, speechiness, valence
Spotify Tracks Dataset (Kaggle)	https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset	CSV	~2,000 tracks with pre-collected audio features, used as supplement
Billboard Hot 100 (Kaggle)	https://www.kaggle.com/datasets/dhruvildave/billboard-the-hot-100-songs	CSV	chart rankings used as additional hit validation
Genius Song Lyrics Dataset (Kaggle)	https://www.kaggle.com/datasets/carlosgdcj/genius-song-lyrics-with-language-information	CSV	word count and language data to enrich the feature set



Data Pipeline & Collection Overview

Step 1: Authentication
- OAuth 2.0 to obtain Spotify API access token

Step 2: Data Collection (Spotify API)
- Collected ~200 hip-hop tracks
- Offset-based pagination due to API limitations

Step 3: Data Limitation
- Spotify API returned popularity = 0 for all tracks
- Unable to directly use API data for hit prediction

Step 4: Data Processing
- Cleaned dataset (removed duplicates, fixed types)
- Extracted release year and converted duration

Step 5: Modeling Data Source
- Used Kaggle datasets (~1,000 tracks) for training
- Provided valid popularity labels for classification

A Dataset Snapshot:

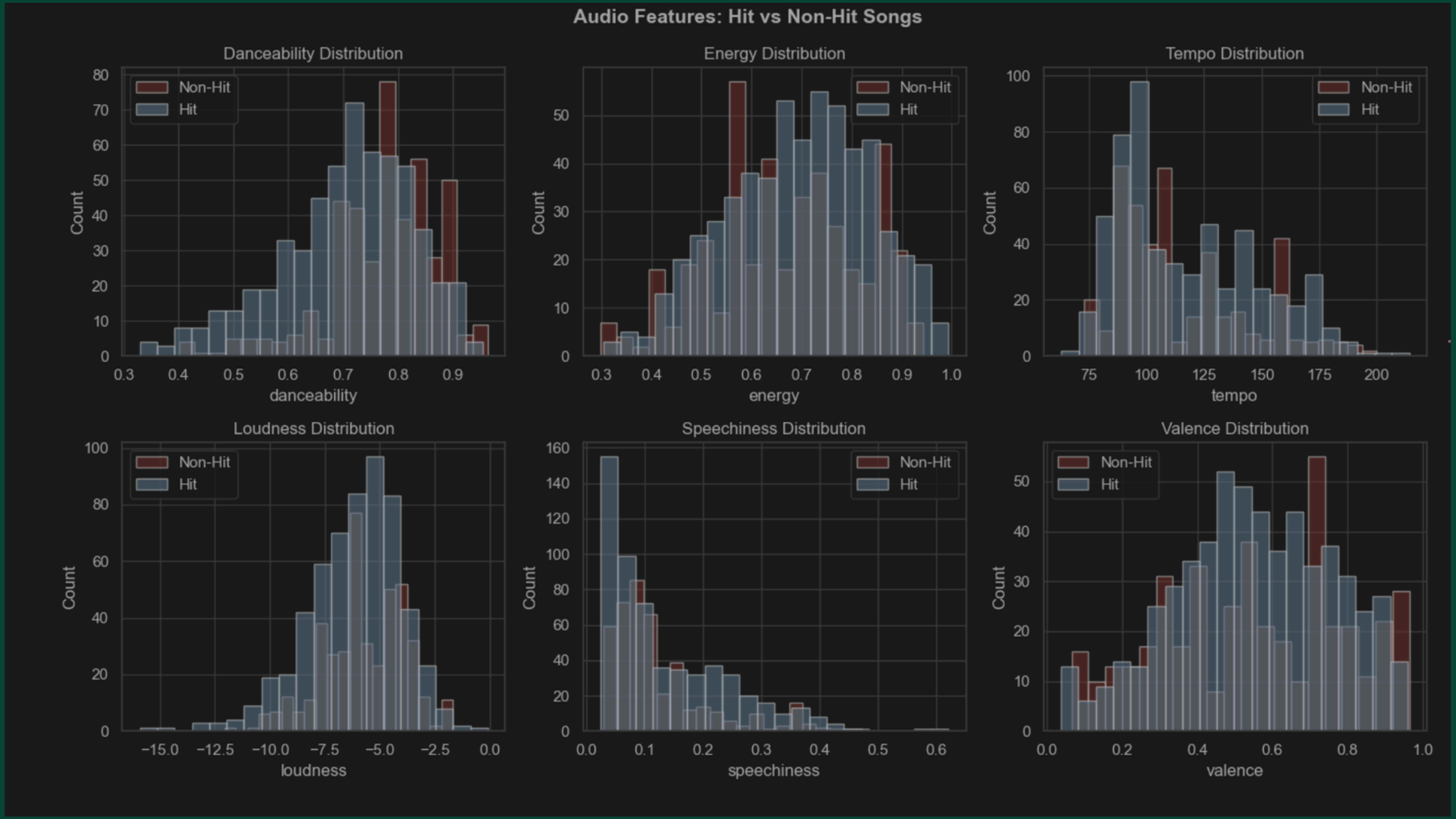
Metric	Value
Tracks collected via Spotify API	**200**
Tracks used for modeling (Kaggle)	**1,000**
Features per track	**10**
Earliest release year	**1988**
Explicit content	**61%**
Hit threshold (popularity)	**≥ 50**

Summary of Audio Features Description

10 audio features used as input variables for the classification models.

Feature	Description	Scale	Type
Danceability	How suitable a track is for dancing based on tempo, rhythm stability, and beat strength. Higher = more danceable.	0.0 — 1.0	Continuous
Energy	Perceptual intensity and activity. Fast, loud, and noisy tracks score higher.	0.0 — 1.0	Continuous
Tempo	Speed of the track in beats per minute (BPM). Hip-hop hits typically range 85—140 BPM.	BPM	Continuous
Loudness	Overall loudness in decibels. Typical range: —60 to 0 dB. Higher (closer to 0) = louder.	dB	Continuous
Speechiness	Presence of spoken words. Higher values indicate more rap or spoken content.	0.0 — 1.0	Continuous
Valence	Musical positiveness. High = happy/euphoric. Low = sad/angry/tense.	0.0 — 1.0	Continuous
Acousticness	Confidence measure of whether the track is acoustic. Higher = more acoustic.	0.0 — 1.0	Continuous
Instrumentalness	Predicts whether a track has no vocals. Values above 0.5 suggest instrumental tracks.	0.0 — 1.0	Continuous
Liveness	Detects presence of a live audience. Higher values suggest the track was performed live.	0.0 — 1.0	Continuous
Explicit	1 if the track contains explicit lyrics, 0 if not. 61% of hip-hop tracks in this dataset are explicit.	0 or 1	Binary

Summary of the result:Audio Feature Distributions Hit vs Non-Hit Songs

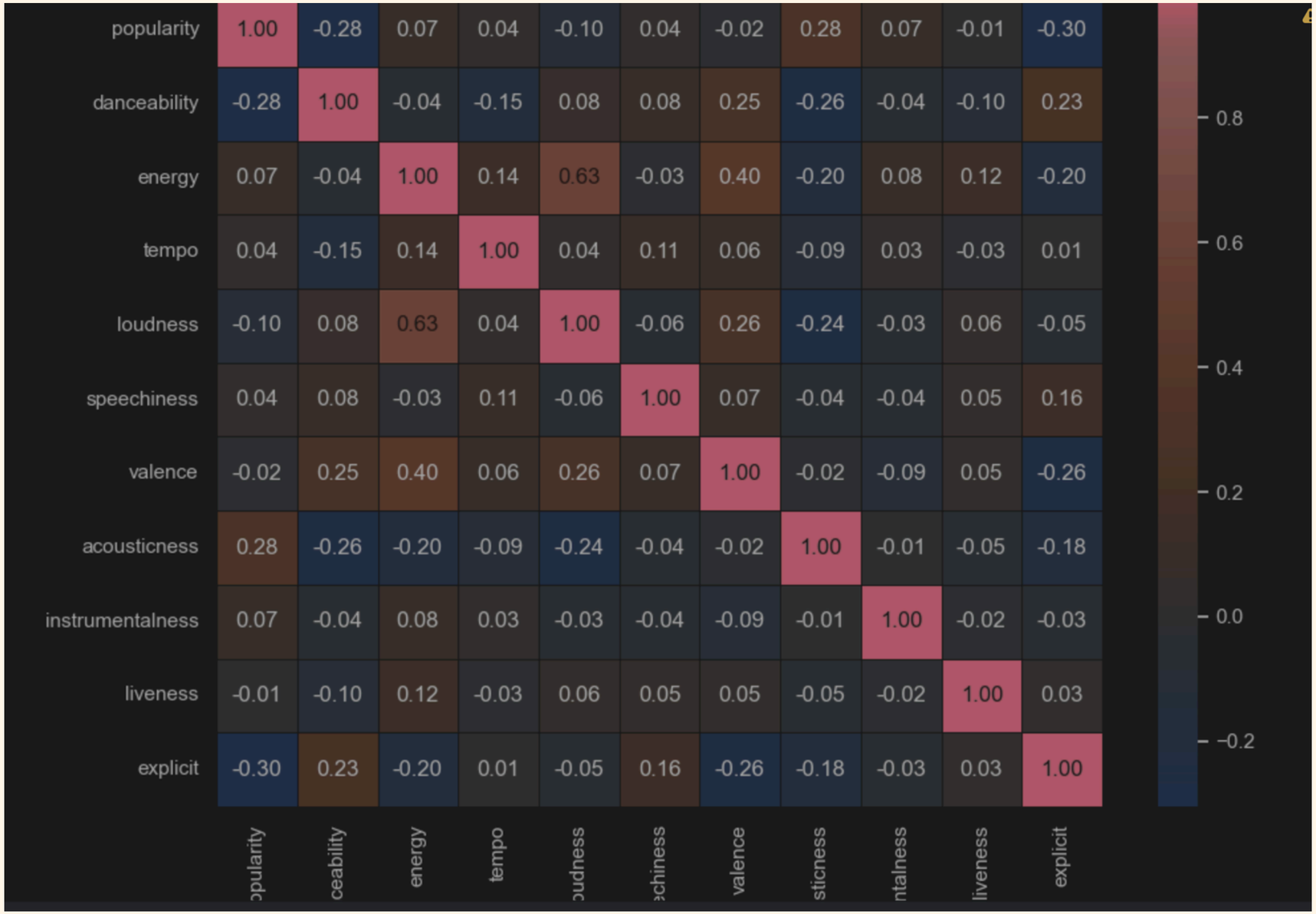


I compares the distribution of each audio features between hit and non-hit songs.

The goal is to identify patterns that may explain musical success.

Result shows hit songs generally have higher danceability, loudness, and valence, suggesting that energetic and emotionally positive tracks are more likely to be popular. Other features, such as speechiness and tempo, show weaker differences.

CorrelationAudio Features and Song Popularity

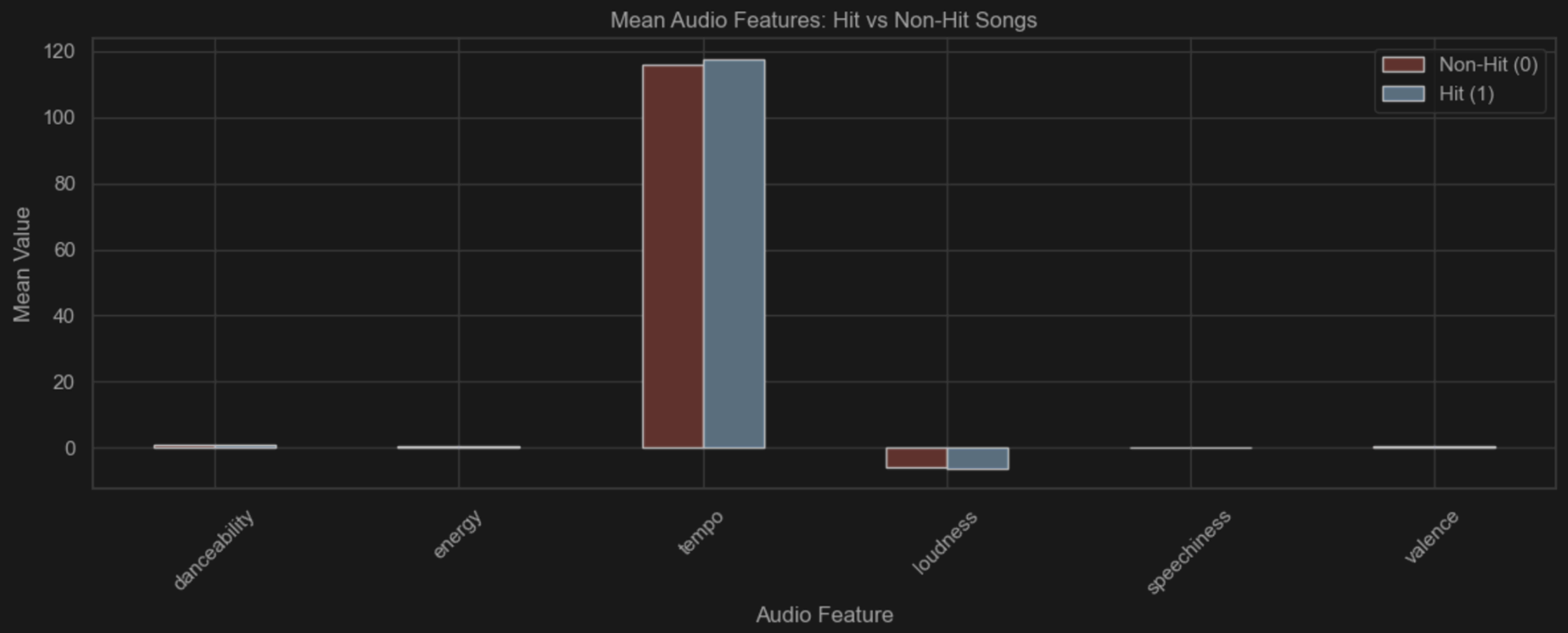


This figure shows the correlation matrix between audio features and song popularity.

Most features have weak individual correlations with popularity, indicating that no single feature can strongly predict success. However, some features, such as acousticness and danceability, show moderate relationships.

Strong correlations between features, such as energy and loudness, suggest that multiple features should be considered together in modeling.

Mean Audio Feature Comparison



is_hit	danceability	energy	tempo	loudness	speechiness	valence
0	0.778	0.667	115.721	-5.745	0.125	0.553
1	0.705	0.694	117.554	-6.158	0.134	0.550

I compares the mean values of audio features between hit and non-hit songs.

Hit songs show slightly higher energy, tempo, and loudness, while differences in other features remain small.

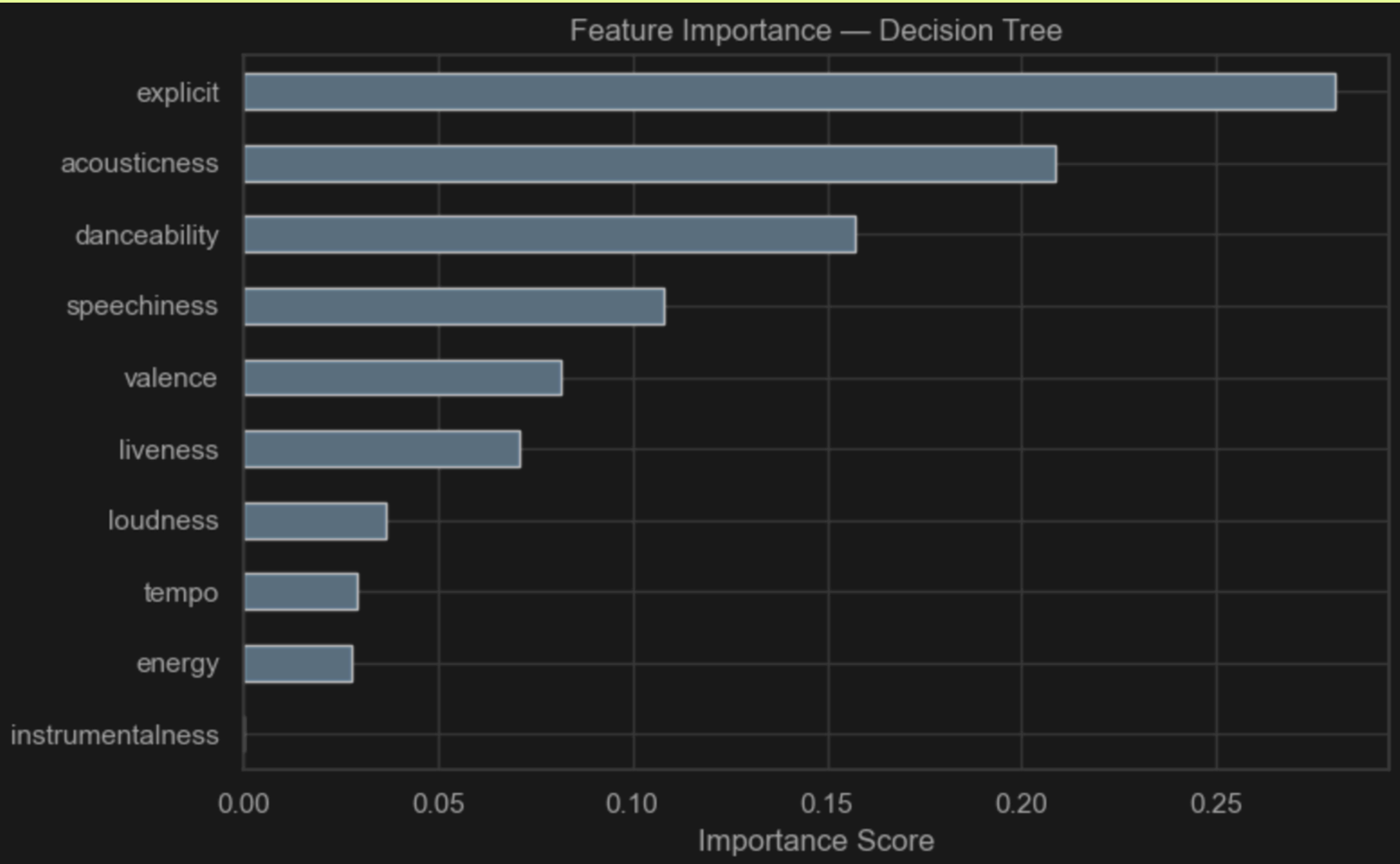
Overall, the differences are not strong enough individually, suggesting that multiple features need to be considered together in predictive modeling.

These observations motivate the use of classification models to better capture combined feature effects.

Feature Importance from Decision Tree Model

The top predictors are explicit content, acousticness, and danceability, indicating that both musical features and content characteristics influence popularity.

Other features contribute less individually, suggesting that multiple factors jointly determine whether a song becomes a hit.

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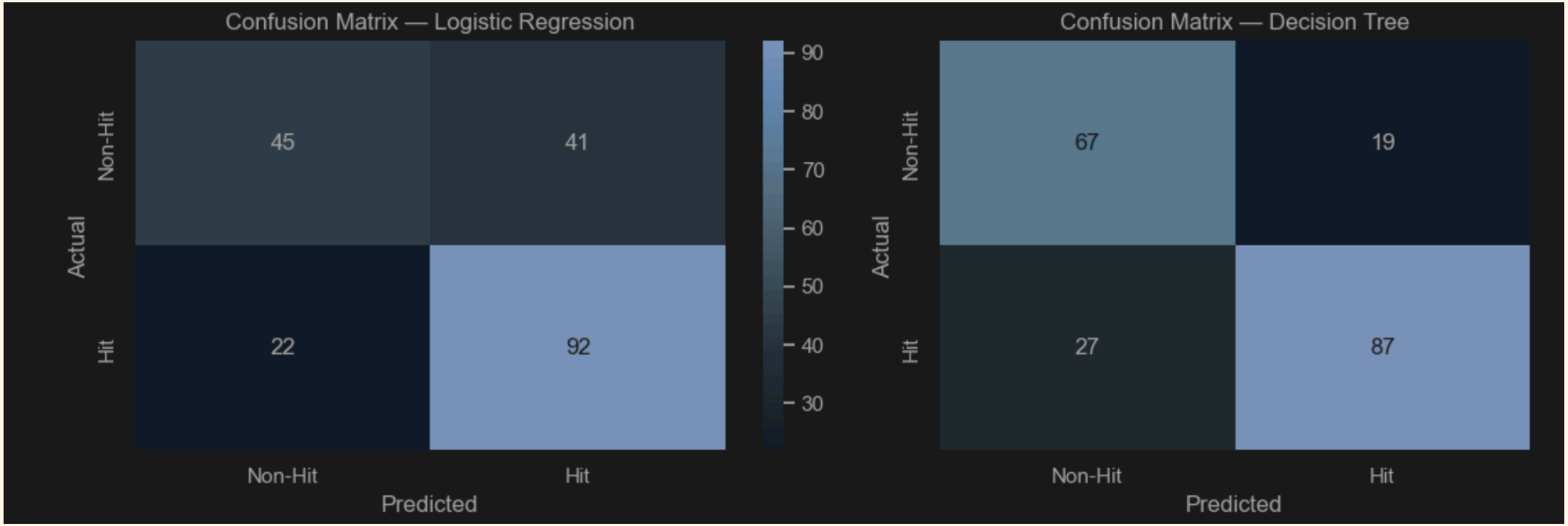
Top 3 most important features:
explicit        0.280410
acousticness    0.208571
danceability    0.157269
dtype: float64
```

I also explains why the Decision Tree model outperforms simpler models by capturing nonlinear relationships between features.

Model Performance Comparison: Logistic Regression vs Decision Tree

Confusion Matrix:Both models were evaluated on 200 unseen test tracks. The Decision Tree correctly identified 154 out of 200 tracks, with particularly high precision — when it predicted a song as a hit, it was right 82% of the time.
In contrast, Logistic Regression shows relatively lower accuracy and makes more misclassifications, especially in distinguishing non-hit songs.



Classification Model Performance Comparison

- Decision Tree: Accuracy = 77%, Precision = 0.82
- Logistic Regression: Accuracy = 69%, Precision = 0.69
- Logistic Regression: Recall = 0.81 (higher than Decision Tree)
- → Captures more actual hits but increases false positives
- Conclusion: Decision Tree provides more reliable predictions

Challenges:

01 — Spotify API Policy Restrictions (Critical)
Spotify's new developer policy zeroed out all popularity scores. All 200 tracks returned popularity = 0, making hit labeling impossible from API data alone.
Workaround: integrate Kaggle datasets as supplementary ground-truth labels.

02 — Rate Limiting & Slow Collection (High)

- API capped at limit=1 per request, requiring individual calls with a 0.3s delay each
- Implemented automatic retry logic; collection was slow but fully functional

03 — Class Imbalance & Model Generalization (Medium)
With 57% hits and 43% non-hits, the dataset is slightly imbalanced. Logistic Regression showed lower accuracy (68.5%) compared to Decision Tree (77%), suggesting audio features alone have limited predictive power and more diverse data sources would improve generalization.

Conclusion

This project shows that while audio features can help predict song popularity, they are not sufficient on their own. The Decision Tree model performs better by capturing more complex relationships, but popularity remains a multifactor outcome rather than the result of a single characteristic.

Through this project, I also learned how to build a complete data pipeline, from API data collection and cleaning to model evaluation and interpretation. More importantly, I gained a deeper understanding of how real-world data limitations, such as missing or restricted features, can significantly impact analysis and require practical adjustments.

Ultimately, understanding music success requires going beyond audio features to include broader contextual and cultural factors.

Thanks for watching!